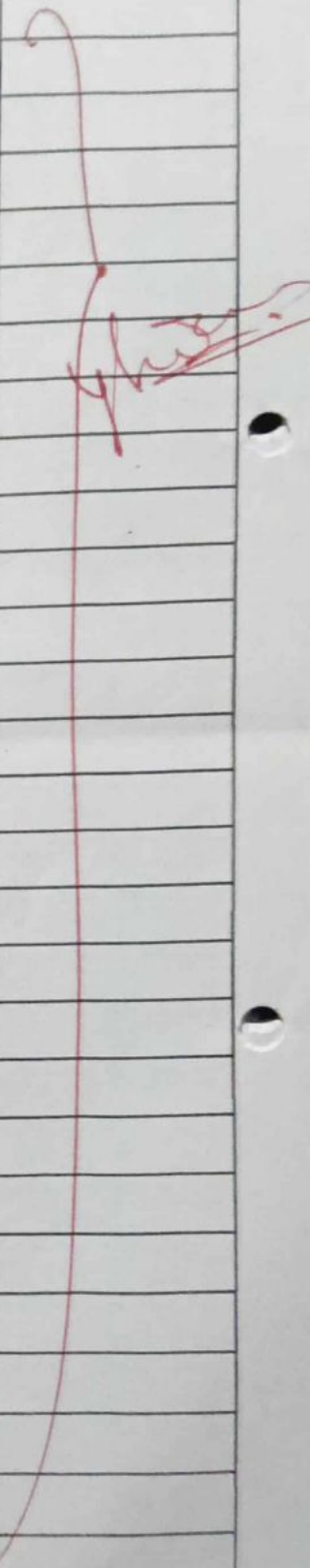


INDEX

SL. NO	TOPICS	SIGN
① a)	Write a python program to print "Hello Python"	
(b)	Write a python program to do arithmetical operations	
②	Write a python program to find the area of triangle	
③	Write a python program to solve quadratic equation	
④	Write a python program to swap two variable	
⑤	Write a python program to convert Celsius to Fahrenheit	
⑥	Write a python program to check if numbers is odd or even.	
⑦	Write a python program to print all prime numbers in an interval	
⑧	Write a python program to find the factorial of a number	
⑨	Write a python program to display the multiplication table	
⑩	Write a python program to multiply two matrices.	
⑪	Write a python program to Find LCM and GCD using functions	
⑫	Write a python program to read a word and print the letters of word, vowels in the word	
⑬	Write a python program to input an array of n numbers and find separately the sum of positive and negative numbers	

INDEX

[illegible]

Lab Program -01

Q. Write a python program to print "Hello Python"

(b) Write a Python Program to do arithmetical operations.

(a) Program to print "Hello python"

```
print("Hello python")
```

Output

Hello python

(b) Arithmetical Operations

```
def add(p, q):
```

```
    return p+q
```

```
def subtract(p, q):
```

```
    return p-q
```

```
def multiply(p, q):
```

```
    return p*q
```

```
def divide(p, q):
```

```
    return p/q
```

```
print("Select operation")
```

```
print("a. Add")
```

```
print("b. Subtract")
```

```
print("c. Multiply")
```

```
print("d. Divide")
```

```
choice = input("enter your choice:")
```

```
num1 = int(input("enter the first no:"))
```

```
num2 = int(input("enter the second no:"))
```

```
if choice == 'a':
```

```
    print(num1, "+", num2, "=", add(num1, num2))
```

```
elif choice == 'b':
```

```
    print(num1, "-", num2, "=", subtract(num1, num2))
```

```
elif choice == 'c':
```

```
    print(num1, "*", num2, "=", multiply(num1, num2))
```

```
elif choice == 'd':
```

```
    print(num1, "/", num2, "=", divide(num1, num2))
```

```
else
```

```
    print("invalid choice")
```

output

Select operation

a. Add

b. subtract

c. multiply

d. divide

enter your choice : d

enter first no: 1

enter second no: 2

1 / 2 = 0.5

Lab Program -02

2) Write a python program to find the area of triangle.

```
base = float(input("enter the base of triangle"))  
height = float(input("enter the height of triangle"))  
area = (base * height) / 2  
print("The area of triangle is", area)
```

Output

enter the base of triangle 6

enter the height of triangle 2

area of triangle is 6

enter the base of triangle 7

enter the height of triangle 2

area of triangle is 7

Lab Program - 03

③ Solve quadratic equations

```
import math
```

```
a = float(input('enter a :'))
```

```
b = float(input('enter b :'))
```

```
c = float(input('enter c :'))
```

```
# calculate the discriminant
```

```
d = (b**2) - (4*a*c)
```

```
# find two solutions
```

```
sol1 = (-b - (math.sqrt(d))) / (2*a)
```

```
sol2 = (-b + (math.sqrt(d))) / (2*a)
```

```
print("The solution are {0} and {1}").
```

```
format(sol1, sol2)
```

Output

enter a : 3

enter b : 5

enter c : 9

Solutions are (-0.8333333333333333-1.518405596522405j) and -0.8333333333333333+1.518405596522405j

Lab Program - 04

4) Write a python program to swap two numbers

```
P = int (input ("enter value for P:"))
```

```
Q = int (input ("enter value for Q:"))
```

```
temp = P
```

```
P = Q
```

```
Q = temp
```

```
print ("The value of P after swapping:", P)
```

```
print ("The value of Q after swapping:", Q)
```

Output

```
enter value P: 12
```

```
enter value Q: 43
```

```
The value of P after swapping  
43
```

```
The value of Q after swapping  
12
```

Lab Program 05

⑤ Write a python program to celsius to celsius to Fahrenheit

```
# convert celsius to Fahrenheit  
celsius = float(input("temperature value in degree  
celsius: "))
```

```
fahrenheit = (celsius * 1.8) + 32  
print("the %.2f degree celsius is equal to : %.2f  
fahrenheit" % (celsius, fahrenheit))
```

convert fahrenheit to celsius

```
fahrenheit = float(input("temperature value in  
degree fahrenheit: "))
```

```
celsius = (fahrenheit - 32) / 1.8
```

```
print("the %.2f degree fahrenheit is equal to :  
%.2f celsius" % (fahrenheit, celsius))
```

Output

temperature value in degree celsius : 32

the 32.00 degree celsius is equal to : 89.60
fahrenheit

temperature value in degree fahrenheit : 94

The 94.00 degree fahrenheit is equal to 34.44
celsius

Lab Program - 06

⑥ Write a python Program to check if numbers is odd or even

```
number = int(input("enter a number"))
```

```
if number % 2 == 0:
```

```
    print("{0} is even".format(number))
```

```
else:
```

```
    print("{0} is odd".format(number))
```

Output

Enter a number 2

2 is even

Lab Program - 07.

⑦ Write a python program to print all prime numbers in an interval.

```
lowervalue = int(input("enter lowervalue"))
```

```
uppervalue = int(input("enter uppervalue"))
```

```
print("prime numbers")
```

```
for number in range(lowervalue, uppervalue + 1)
```

```
    if number > 1:
```

```
        for i in range(2, number):
```

```
            if (number % i) == 0:
```

```
                break
```

```
        else:
```

```
            print(number)
```

Output

enter lowervalue 6

enter upper value 10

Prime numbers 7

Lab Program - 08

8) write a python program to find the factorial of a number.

```
num = int(input("enter a number"))
```

```
fact = 1
```

```
if num < 0:
```

```
    print("the factorial doesnot exist for negative numbers")
```

```
elif num == 0:
```

```
    print("the factorial of 0 is 1")
```

```
else:
```

```
    for i in range(1, num+1):
```

```
        fact = fact * i
```

```
    print("The factorial of given no is ", fact)
```

output

enter a number 5

The factorial of given
number is 120

Lab Program - 09

⑨ Write a python program to display the multiplication table.

```
number = int(input("enter the number to print the  
multiplication table:"))
```

```
count = 1
```

```
#while loop for iterating the multiplication 10 times
```

```
print("The multiplication table of: ", number)
```

```
while count <= 10:
```

```
    number = number * 1
```

```
    print(number * count)
```

```
    count += 1
```

Output

Enter the number to print the multiplication
table : 10

The multiplication Table of : 10

10

20

30

40

50

60

70

80

90

Lab Program -10

⑩ Write a python program to multiply two matrix

```
import numpy as np

# Define two matrix
A = np.array([[1, 2],
               [3, 4]])

B = np.array([[5, 6],
               [7, 8]])

# Multiply the matrix
result = np.dot(A, B)

# Print the result
print("Matrix A:")
print(A)

print("Matrix B:")
print(B)

print("Product A and B:")
print(result)
```

Output

Matrix A:

```
[[1 2]
 [3 4]]
```

Matrix B:

```
[[5 6]
 [7 8]]
```

Product A and B:

```
[[19 22]
 [43 50]]
```

Lab Program - 11

① Write a python Program to Find LCM & GCD using function

```
def g(a,b):  
    gcd = 1  
    for i in range(1, a+1):  
        if a%i == 0 and b%i == 0:  
            gcd = i  
    return gcd.
```

```
first = int(input("enter the first number:"))  
second = int(input("enter second number:"))  
print("GCD of %d and %d is %d" % (first, second, gcd))
```

Output

Enter first number: 3

Enter second number: 4

HCF or GCD of 3 and 4 is 1

LCM of 3 and 4 is 12

Lab Program -12

12) Write a python program to read a word and print the number of letters, vowels in the word

```
string = input("enter a string:")
```

```
vowels = ['a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U']
```

```
tot1 = 0
```

```
tot2 = 0
```

```
for line in string:
```

```
    print(line)
```

```
for char in string:
```

```
    if char >= 'a' and char <= 'z':
```

```
        if char not in vowels:
```

```
            tot2 = tot2 + 1
```

```
    elif char >= 'A' and char <= 'Z':
```

```
        if char not in vowels:
```

```
            tot2 = tot2 + 1
```

```
print("Number of consonant are:")
```

```
print(tot2)
```

```
for char in string:
```

```
    if char in vowels:
```

```
        tot1 = tot1 + 1
```

```
print("Number of vowels are:")
```

```
print(tot1)
```

Output

enter a string : Hello world

H

e

l

l

o

W

o

r

i

d

Number of consonant are:

7

Number of vowels are:

3

Lab Program -13

- ⑬ Write a python program to input an array of n numbers and find separately the sum of positive numbers and negative numbers

```
def sep(numbers):
```

```
    pos = 0
```

```
    neg = 0
```

```
    for num in numbers:
```

```
        if num > 0:
```

```
            pos += num
```

```
        elif num < 0:
```

```
            neg += num
```

```
    return pos, neg
```

```
n = int(input("enter the length of array"))
```

```
numbers = []
```

```
for i in range(n):
```

```
    num = int(input("enter the number"))
```

```
    numbers.append(num)
```

```
pos, neg = sep(numbers)
```

```
print("sum of positive number is", pos)
```

```
print("sum of negative number is", neg)
```

Output

enter the length of array 2

enter the number 2

enter the number 5

Sum of positive number 7

Sum of negative number 0

Lab Program -14

- (14) Write a python program to search an element using linear equation implement a sequential search.

```
def linear-search(alist, key):  
    for i in range(len(alist)):  
        if alist[i] == key:  
            return i  
    return -1
```

```
alist = input("enter the list of numbers:")  
alist = alist.split()  
alist = [int(x) for x in alist]  
key = int(input("The number to search for:"))  
index = linear-search(alist, key)  
if index < 0:  
    print('{} was not found.'.format(key))  
else:  
    print('{} was found at index {}'.format(  
        key, index))
```

Output

```
Enter the list of numbers 6 7 8 9 5 4 3 9 9  
The number to be search for: 3  
3 was found at index 6
```


Lab Program - 15

⑤ Write a python program to search an element using binary search

```
def binary-search(arr, target):
```

```
    low = 0
```

```
    high = len(arr) - 1
```

```
    while low <= high:
```

```
        mid = (low + high) // 2
```

```
        if arr[mid] == target:
```

```
            return mid    # Element found at index mid
```

```
        elif arr[mid] < target:
```

```
            low = mid + 1
```

```
        else:
```

```
            high = mid - 1
```

```
    return -1    # Element not found
```

```
arr = [1, 3, 5, 7, 9, 11, 13, 15]
```

```
target = int(input("Enter the element to search:"))
```

```
result = binary-search(arr, target)
```

```
if result != -1:
```

```
    print("Element found at index:", result)
```

```
else:
```

```
    print("Element not found in the list.")
```

Lab Program - 15

Output

enter the element to search 9

Element is found at 4 index

Output

enter the element to search 18

Element not found in the list

Lab Program -16

⑩ Write a Python program to insert a number in sorted array

```
def insert(list, n):  
    list.append(n)  
    sorted_list = sorted(list)  
    return sorted_list  
  
list = [1, 2, 3]  
n = 3  
print(insert(list, n))
```

Output

[1, 2, 3, 3]

Lab Program -17

⑦ Write a python program to stimulate stack operation.

```
stack = []
```

```
stack.append('a')
```

```
stack.append('b')
```

```
stack.append('c')
```

```
print('initial stack')
```

```
print(stack)
```

```
print('element deleted from stack')
```

```
print(stack.pop())
```

```
print('after stack.pop()')
```

```
print('after element are popped')
```

```
print(stack)
```

Output

initial stack

['a', 'b', 'c']

element deleted from stack

c

b

a

after element are popped

- 18) Write a python program to stimulate stack operation to draw shapes & GUI controls

```

from tkinter import *
from functools import partial

def validateLogin(username, password):
    print("username entered:", username.get())
    print("password entered:", password.get())
    return

t = Tk()
t.geometry('400x150')
t.title('Tkinter Login Page')
usernameLabel = Label(t, text = "User Name").grid(
    row=0, column=0)

username = StringVar()
usernameEntry = Entry(t, textvariable = username).grid(
    row=0, column=1)

passwordLabel = Label(t, text = "Password").grid(
    row=1, column=0)

password = StringVar()
passwordEntry = Entry(t, textvariable = password, show = '*')
    .grid(row=1, column=1)

validateLogin = partial(validateLogin, username,
    password)

```

Lab Program -18

```
login_Button = Button(t, text = "login", command = validate  
Login).grid(row=4, column=0)
```

```
t.mainloop()
```

Output

Username entered : Alice

Password entered : 1234

Lab Program -19

19) Write a python programs to using the built in Methods of the string, list, dictionary

Built in methods of the string

```
a = 'hi'
```

```
a.capitalize()
```

```
>>> 'Hi'
```

```
a = 'hi'
```

```
a.upper()
```

```
>>> 'HI'
```

```
a = 'HI'
```

```
a.lower()
```

```
>>> 'hi'
```

```
a = 'hi'
```

```
a.title()
```

```
>>> 'Hi'
```

```
a = 'hi'
```

```
a.swapcase()
```

```
>>> 'HI'
```

```
a = 'Happy Birthday'
```

```
a.split()
```

```
>>> ['Happy', 'Birthday']
```

a = 'Happy Birthday'

a.center(18, '*')

>>> '* Happy Birthday *'

a = 'Happy Birthday'

a.count('Birthday')

>>> 1

a = 'Happy Birthday'

a.replace('Happy', 'wishyou happy')

>>> 'wishyou happy Birthday'

a = 'Happy Birthday'

b = "happy"

a.join(b)

>>> 'hHappy BirthdayaHappy BirthdaypHappy
BirthdaypHappy Birthdayy'

Built in methods in the list

Numbers = [2, 4, 6]

Numbers.append(3)

print("after appending", Numbers)

>>> [2, 4, 6, 3]


```
languages = ['kannada', 'english']
```

```
lang = ['Tamil', 'Hindi']
```

```
languages.extend(lang)
```

```
print('Language list is', languages)
```

```
>>> ['kannada', 'english', 'Tamil', 'Hindi']
```

```
lang = ['kannada', 'english']
```

```
lang.insert(2, 'Hindi')
```

```
print('Language is', lang)
```

```
>>> ['kannada', 'english', 'Hindi']
```

```
lang = ['python', 'java']
```

```
lang.pop()
```

```
print(lang)
```

```
>>> ['python']
```

```
cities = ["oslo", "delhi", "washington", "london",  
          'seattle', "paris", "washington"]
```

```
cities.count('seattle')
```

```
>>> 1
```

Built in methods of dictionary

```
a = {'m', 'b', 'v'}
```

```
n = 1
```

```
d = dict.fromkeys(a, n)
```

```
print(d)
```

```
>>> {'b': 1, 'm': 1, 'v': 1}
```

```
Names = {1: 'Abhi', 2: 'Raja', 3: 'Rakesh'}
```

```
Names.pop('1')
```

```
print(Names)
```

```
>>> {2: 'Raja', 3: 'Rakesh'}
```

```
Names = {1: 'Abhi', 2: 'Raja', 3: 'Rakesh'}
```

```
Names.clear()
```

```
print(Names)
```

```
>>> {}
```

```
Names = {1: 'Abhi', 2: 'Raja', 3: 'Rakesh'}
```

```
Names.get('2')
```

```
print(Names)
```

```
>>> Raja
```

```
Numbers = {'w': 4, 'e': 2, 'r': 12, 't': 9}
```

```
sorted_no = sorted(Numbers)
```

```
print(sorted_no)
```

```
>>> 'e', 'r', 't', 'w'
```


Lab Program - 20.

(20) Write a python program to demonstrate exception handling

```
try:
```

```
    temp = [1, 2, 3]
```

```
    temp[3]
```

```
except Exception as e:
```

```
    print("Exception block handling: ", e)
```

```
else:
```

```
    print("No exception found")
```

```
finally:
```

```
    print("finally block executed")
```

Output

Exception block handling

finally block executed

Seen
6/6/25